

AI ASSISTED CODING

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BATCH – 03

10 – 03 – 2026

ASSIGNMENT – 13.2

LAB – 13 : Code Refactoring – Improving Legacy Code with AI Suggestions.

Task – 01 : Refactoring – Eliminating Repetitive Logic.

Prompt : Use AI to analyze the Python code and identify repeated calculations for area and perimeter. Refactor the code by creating a reusable function with proper docstrings while ensuring the output remains the same.

Original Code & Output :



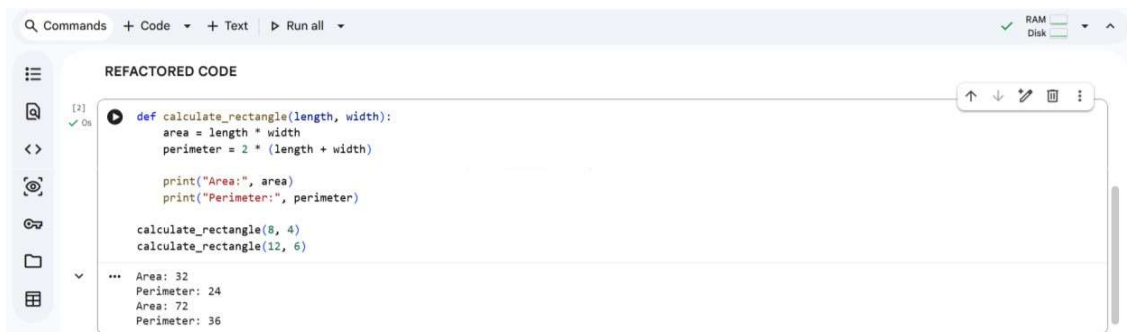
The screenshot shows a code editor interface with a sidebar on the left containing icons for file explorer, search, and other tools. The main area is titled 'TASK 01' and 'ORIGINAL CODE'. It contains two blocks of Python code. The first block calculates area and perimeter for length=8 and width=4. The second block calculates area and perimeter for length=12 and width=6. Below the code, the output is shown: 'Area: 32', 'Perimeter: 24' for the first case, and 'Area: 72', 'Perimeter: 36' for the second case. The bottom status bar shows 'Variables', 'Terminal', '1:30 PM', and 'Python 3'.

```
length = 8
width = 4
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))

length = 12
width = 6
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))
```

Area: 32
Perimeter: 24
Area: 72
Perimeter: 36

Refactored Code & Output :



The screenshot shows the same code editor interface, but now titled 'REFACTORED CODE'. It contains a single Python function 'calculate_rectangle' that takes 'length' and 'width' as arguments and returns the area and perimeter. Below the function, the function is called with the same two sets of input values as in the original code. The output is identical to the original code: 'Area: 32', 'Perimeter: 24' and 'Area: 72', 'Perimeter: 36'.

```
def calculate_rectangle(length, width):
    area = length * width
    perimeter = 2 * (length + width)

    print("Area:", area)
    print("Perimeter:", perimeter)

calculate_rectangle(8, 4)
calculate_rectangle(12, 6)
```

Area: 32
Perimeter: 24
Area: 72
Perimeter: 36

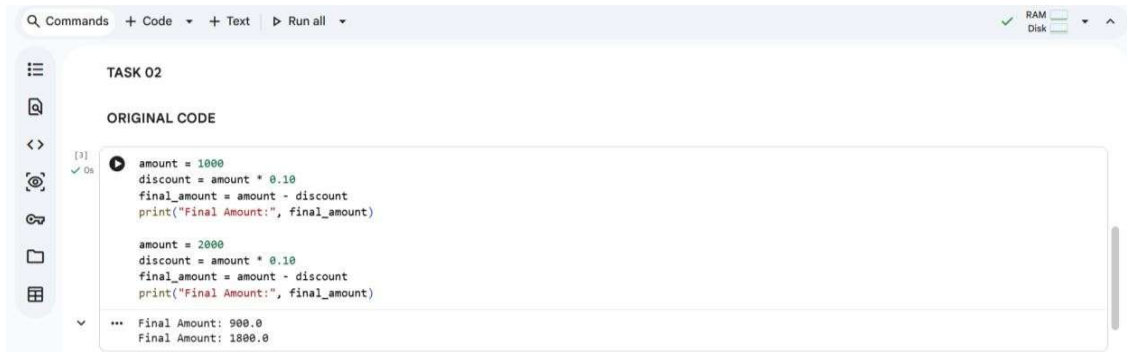
Explanation :

The repeated calculations for area and perimeter were moved into a reusable function called `calculate_rectangle`. This improves code readability and reduces duplication. The program still produces the same output but is easier to maintain.

Task – 02 : Refactoring – Modularizing Inline Calculations.

Prompt : Use AI to refactor the Python code by identifying repeated discount calculations and converting them into a reusable function with proper documentation.

Original Code & Output :



```
TASK 02

ORIGINAL CODE

[3] ✓ Os
amount = 1000
discount = amount * 0.10
final_amount = amount - discount
print("Final Amount:", final_amount)

amount = 2000
discount = amount * 0.10
final_amount = amount - discount
print("Final Amount:", final_amount)

*** Final Amount: 900.0
    Final Amount: 1800.0
```

Refactored Code & Output :



```
REFACTORED CODE

[4] ✓ Os
def calculate_final_amount(amount):
    discount = amount * 0.10
    final_amount = amount - discount
    print("Final Amount:", final_amount)

calculate_final_amount(1000)
calculate_final_amount(2000)

*** Final Amount: 900.0
    Final Amount: 1800.0
```

Explanation :

The repeated discount calculation logic was moved into a function called `calculate_final_amount`. This makes the code modular and reusable for different input values. It also improves readability and reduces redundant code.

Task – 03 : Refactoring – Improving Loop and Conditional Performance.

Prompt : Use AI to optimize the Python program by replacing inefficient nested loops with a faster lookup structure while keeping the output unchanged.

Original Code & Output :



The screenshot shows a code editor interface with a sidebar on the left containing icons for file management and search. The main area is titled 'TASK 03' and 'ORIGINAL CODE'. The code defines a list of students and a list of names to check. It uses nested loops to check if each name is in the students list. The output shows 'Bhavya found'.

```
students = ["Anil", "Bhavya", "Charan", "Divya"]
check_list = ["Charan", "Esha", "Bhavya"]
for name in check_list:
    exists = False
    for student in students:
        if name == student:
            exists = True
    if exists:
        print(name, "found")
    else:
        print(name, "not found")
```

... Bhavya found

Refactored Code & Output :



The screenshot shows the same code editor interface but with the 'REFACTORED CODE'. The code uses a set to store the students list, which allows for faster membership checks using the 'in' operator. The output shows 'Charan found', 'Esha not found', and 'Bhavya found'.

```
students = ["Anil", "Bhavya", "Charan", "Divya"]
check_list = ["Charan", "Esha", "Bhavya"]

student_set = set(students)

for name in check_list:
    if name in student_set:
        print(name, "found")
    else:
        print(name, "not found")
```

... Charan found
Esha not found
Bhavya found

Explanation :

The nested loops were replaced with a set data structure, which provides faster lookup operations. Instead of checking each student repeatedly, membership is checked directly using `in`. This improves performance from $O(n^2)$ to $O(n)$.

Task – 04 : Refactoring – Replacing Hardcoded Values with Constants.

Prompt : Use AI to identify hardcoded numerical values in the simple interest calculation and replace them with descriptive constants to improve maintainability.

Original/Refactored Code & Output :

```
Q Commands + Code + Text ▶ Run all ✓ RAM Disk
```

TASK 04

ORIGINAL CODE

```
[7] ✓ Os  
print("Simple Interest:", (1000 * 2 * 5) / 100)  
print("Simple Interest:", (2000 * 2 * 5) / 100)  
Simple Interest: 100.0  
Simple Interest: 200.0
```

REFACTORED CODE

```
[8] ✓ Os  
RATE = 2  
TIME = 5  
PERCENT = 100  
def calculate_simple_interest(principal):  
    interest = (principal * RATE * TIME) / PERCENT  
    print("Simple Interest:", interest)  
    calculate_simple_interest(1000)  
    calculate_simple_interest(2000)  
Simple Interest: 100.0  
Simple Interest: 200.0
```

Variables Terminal 1:48 PM Python 3

Explanation :

Hardcoded values such as rate, time, and percentage were replaced with named constants (RATE, TIME, PERCENT). This makes the code easier to understand and modify. If the rate or time changes, it only needs to be updated once.

Task – 05 : Refactoring – Enhancing Variable Naming Code Clarity.

Prompt : Use AI to improve code readability by replacing unclear variable names with meaningful ones and adding comments explaining the logic.

Original/Refactored Code & Output :

```
Q Commands + Code + Text ▶ Run all ✓ RAM Disk
```

ORIGINAL CODE

```
[9] ✓ Os  
x = 50  
y = 30  
z = x + y  
print(z)  
80
```

REFACTORED CODE

```
[10] ✓ Os  
# Define two numbers  
num1 = 50  
num2 = 30  
  
# Calculate the sum of the numbers  
total_sum = num1 + num2  
  
# Display the result  
print(total_sum)  
80
```

Variables Terminal 1:52 PM Python 3

Explanation :

The variables x, y, and z were renamed to num1, num2, and total_sum to improve clarity. Comments were added to explain each step of the code. This makes the program easier to understand for other developers.

THANK YOU!!